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Experiment – 1

Name: Parit Yadav

UID: 25BAI70936

Branch: CSE – AIML

Section/Group: 13 / B

Semester: 3rd

Date of Performance: 01/08 / 2026

Subject: Agile Development Methodologies

Subject Code: 25CSH-233

EXPERIMENT – 1.1

1. **AIM:** Create an Empathy Map and As-is Scenario Map for Cab Booking using MURAL.

2. **OBJECTIVE:**

- To identify user pain points, needs, and emotions during the cab booking process.
- To create an Empathy Map to visualize user feelings, thoughts, and behaviours.
- To develop an As-Is Scenario Map to illustrate the existing step and interactions.
- To use MURAL as a digital collaboration tool for designing user – centric maps.

3. **SOFTWARE REQUIREMENTS:**

- MURAL

4. **THEORY**

Empathy Map: An Empathy Map is a strategic visualization tool used in Design Thinking to synthesize qualitative research data into a coherent profile of a specific user persona. It helps teams establish a shared understanding of user needs, attitudes, and behaviours.

The framework is divided into four distinct quadrants anchored around a central persona:

- **Says:** Direct quotes or statements made by the user during research (e.g., verbal complaints or requests).
- **Thinks:** Internal monologues, beliefs, or unstated worries that govern decision-making.
- **Does:** Observable physical or digital interactions (e.g., tapping screens, switching apps, checking timestamps).

- **Feels:** Internal emotional states experienced throughout the interaction lifecycle (e.g., frustration, anxiety, relief).

As-Is Scenario Map: An As-Is Scenario Map is a sequential blueprint of a user's existing end-to-end journey when attempting to achieve a specific goal using a system or service. Unlike idealized workflows, the As-Is map explicitly models the current state of the product, including its systematic flaws, inefficiencies, and edge cases.

The structure relies on a linear progression composed of:

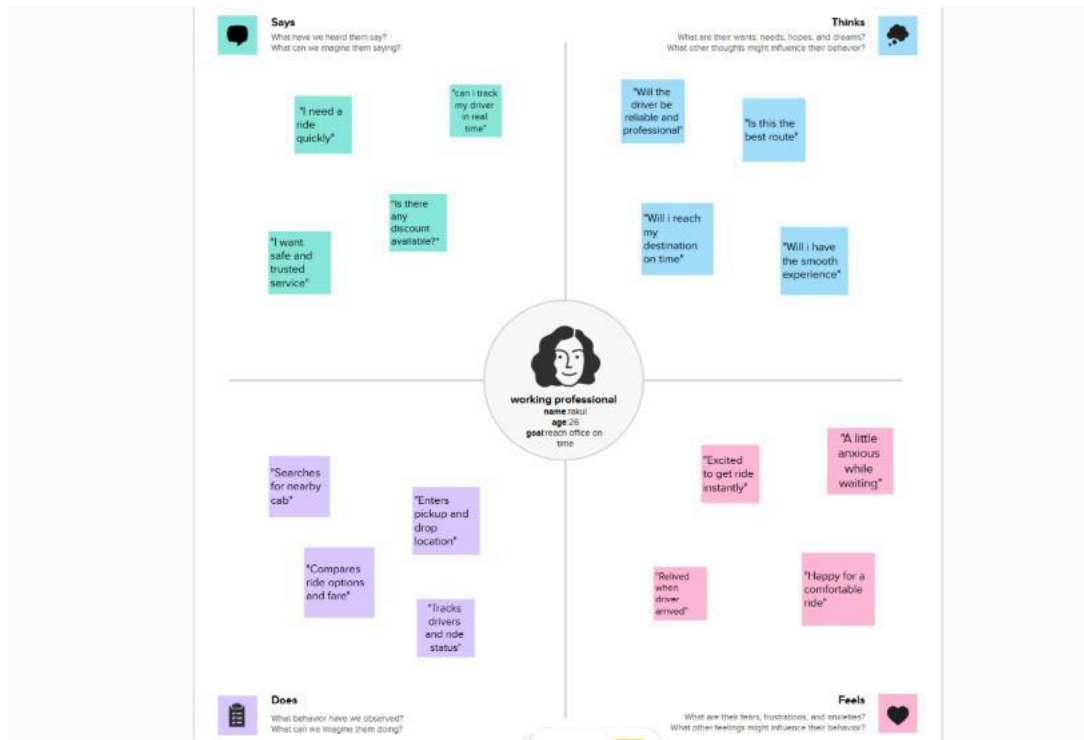
- **Phases & User Steps:** Chronological milestones in the user journey (e.g., App Initiation, Booking Confirmation, In-Transit Phase, Post-Trip Payment).
- **User Actions:** Specific touchpoints and mechanical inputs executed by the user.
- **App Touchpoints & Tools:** UI components, backend integrations, and platform capabilities involved at each step (e.g., GPS modules, payment gateways).
- **Pain Points & Bottlenecks:** System failures, delayed response times, UI confusion, or operational friction (e.g., driver cancellations, inaccurate ETAs).
- **Emotional Trajectory:** The fluctuation of user sentiment across the workflow stages.

5. IMPLEMENTATION/PROCEDURE:

(A) Empathy Map Creation:

1. Open MURAL and start a new blank board or use a pre-designed Empathy Map template.
2. Identify the use persona.
3. Divide the board into four quadrants.
 - **Says:** What the user verbally express (e.g. I need to reach on time).
 - **Thinks:** What the user is thinking but may not say (e.g. What if the driver cancels).

- **Does:** The user's observable actions (e.g. Checking fare, tracking driver). □ **Feels:** Emotions experienced (e.g. anxious, relieved, irritated).
4. Use sticky notes to fill in insights gathered through user interviews, surveys, or personal observation.



(B) Scenario Map Creation

1. In a new section of the MURAL board, create a timeline or flowchart to illustrate the cab booking journey.
2. Identify and document each user step in the process:
 - Open the app
 - Enter pickup and drop-off locations ○ View and compare fare estimates ○ Select the type of cab ○ Confirm and book the ride ○ Ride in the cab ○ Make payment ○ Provide

feedback or rating 3. For each step,
include:

- Actions taken by the user ○
- Tools or app features used ○
- Pain points or obstacles faced ○
- Emotional response or reaction

4. Highlight any significant issues such as driver cancellations, GPS errors, or fare mismatches.

Phases	Discover & Explore	Book Ride	During Ride	After Ride
Doing	Browses cab options Checks fare estimate	Enters pickup & drop details Chooses cab type	Tracks driver in real-time Communicates if needed	Makes payment online Gives feedback or rating
Thinking	Is this the best option? Are there any hidden charges?	Will I get a driver quickly? Is the fare reasonable?	Will I reach on time? Is the route safe?	Was the service worth the price? Will I use this again?
Feeling	Curious Hopeful	Excited Slightly nervous	Comfortable Relieved	Satisfied Appreciated

6. Result

The Empathy Map and As-Is Scenario Map for the cab booking process were successfully created on MURAL. The Empathy Map captured what users think, feel, do, and say, while the As-Is Scenario Map highlighted current challenges such as unreliable ETAs, driver cancellations, and confusion during pickup. These maps provide a strong foundation for redesigning an improved, user-centered cab booking experience.

7. LEARNING OUTCOME

- Understood how to apply empathy in user research through mapping exercises
- Learned how to build an Empathy Map and As-Is Scenario Map using MURAL
- Gained insights into common user frustrations and expectations in digital cab services
- Identified areas of improvement in the cab booking process using a structured visual framework



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Experiment – 2

Name: Parit yadav

UID: 25BAI70936

Branch: CSE – AIML

Section/Group: 13 / B

Semester: 3rd

Date of Performance: 12 / 08 / 2026

Subject: Agile Development Methodologies

Subject Code: 25CSH-233

Exp 1.2

Aim of the Experiment:

Create Big Idea Vignettes, Prioritization Grid and Need Statements for case discussed in Practical 1.

Objective:

1. To generate creative solution ideas (Big Idea Vignettes) based on insights from the Empathy Map and As-Is Scenario
2. To evaluate and organize these ideas using a Prioritization Grid on the basis of user value and implementation effort
3. To formulate clear and actionable Need Statements to guide problem-solving and design
4. To visualize all elements using MURAL for better team collaboration and design iteration

Software Required: Mural

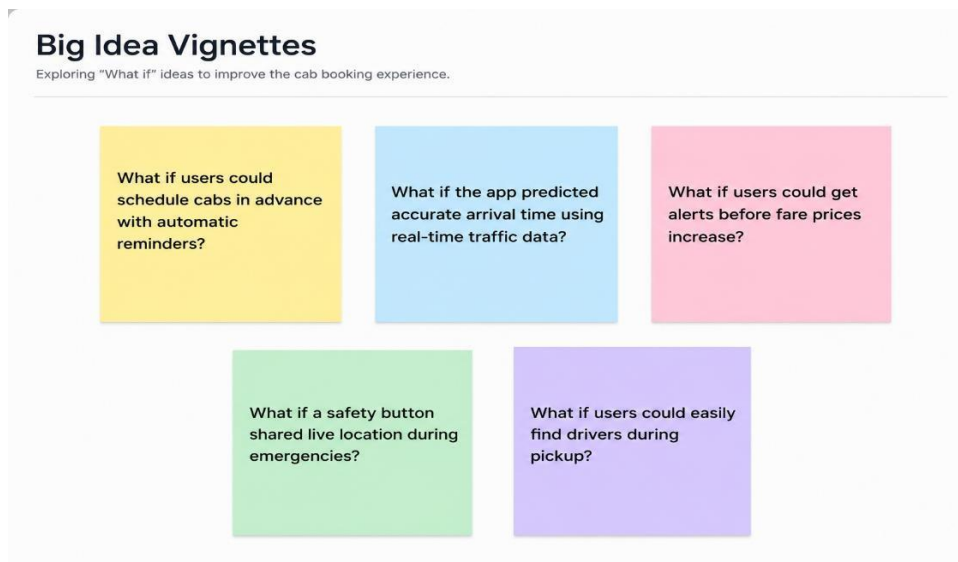
Steps of Procedure:

A. Big Idea Vignettes

1. Revisit the Empathy Map and As-Is Scenario created during Practical 1, identifying major pain points, user expectations, and emotional feedback
2. Based on these insights, write 3–5 Big Idea Vignettes, which are short, imaginative statements that present a new way of solving a key user problem
3. Use the “What if...” format to encourage open-ended thinking 4. Example vignettes: ○ “What if users could schedule cabs in advance with automated reminders before departure?” ○ “What if the app provided a panic button that instantly shared live location with emergency contacts and the police?”

- “What if a dynamic pricing tracker helped users choose the most cost-effective time slot to book a cab?”

5. Add these vignettes in the MURAL board using sticky notes under the section ‘Big Idea Vignettes’

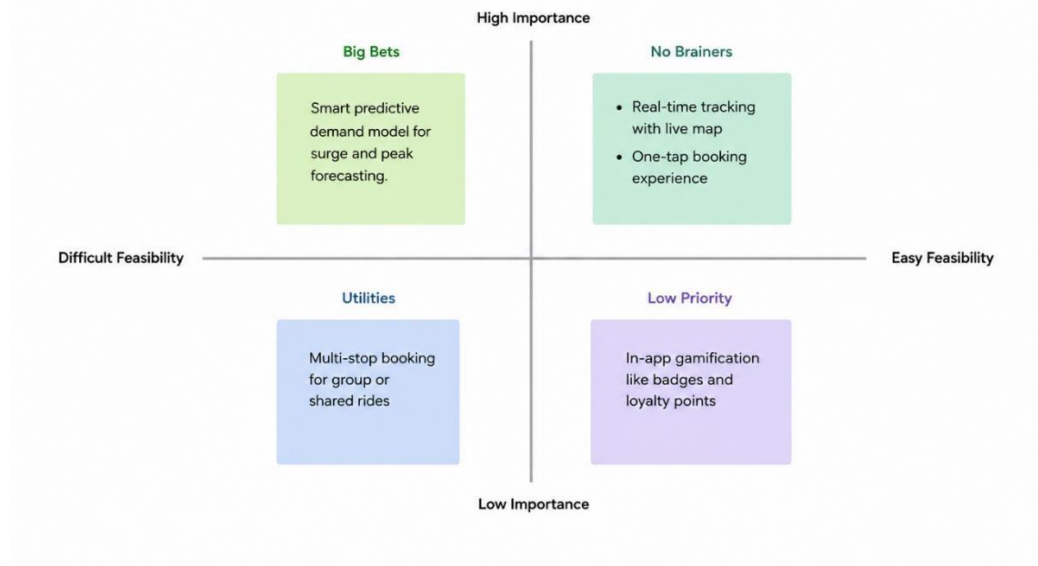


B. Prioritization Grid

1. Come prepared with ideas Prioritization is only as valuable as the ideas you bring to the table, so make sure you have a strong set of ideas to prioritize (for example, from a Big-Idea Vignettes activity or User Stories in your backlog).
2. Set up the activity On your whiteboard, large sticky pad, or virtual whiteboard, draw two axes: “Importance to the user” (low to high) and “Feasibility for the team” (difficult to easy).
3. Evaluate ideas Have everyone quickly evaluate each idea on their own, and roughly plot them on the grid where they make sense. Once many items are on the grid, start discussing with your teammates and reposition them in relation to each other.
4. Focus the discussions Label the upper right-hand quadrant “No Brainers,” the center-right area “Utilities,” and the center-top area “Big Bets

Prioritization Grid

Evaluate ideas based on importance and feasibility to identify what to focus on next.



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B. Need Statements

1. Convert major pain points from Practical 1 into well-structured Need Statements that describe user problems clearly and empathetically
2. Use the format: [User persona] needs a way to [task or solution] because [insight or reason] 3.
Example Need Statements: ○ “A solo traveler needs a way to ensure their cab is secure because they often travel late at night and feel unsafe.”
 - “A budget-conscious user needs a way to avoid surge pricing because they rely on cabs for their daily commute.”
4. Add these statements to MURAL under the ‘Need Statements’ section using distinct color coding or labels

Need Statements

Frame key user needs based on insights from the empathy and scenario mapping.

A working **professional** needs a way to reach the office on time because delays affect work.

A commuter needs a way to **avoid driver cancellations** because they waste time.

A user needs a way to know the **correct fare** because price changes cause frustration.

A traveler needs a way to **feel safe** because they may travel alone.

A user needs a way to find the **pickup location easily** because GPS can be confusing.

Result:

Three distinct outputs were generated based on the case study: Big Idea Vignettes that reimaged the cab booking experience, a Prioritization Grid that helped rank ideas based on impact and feasibility, and Need Statements that captured core user requirements. These deliverables form the basis for user-centered design ideation and solution development.

Learning Outcomes:

1. Understood how to translate user insights into visionary solution concepts
2. Practiced evaluating solution ideas based on usability and feasibility metrics
3. Developed clarity in framing actionable and user-specific Need Statements



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Experiment – 3

Name: Parit Yadav

UID: 25BAI70936

Branch: CSE – AIML

Section/Group: 13 / B

Semester: 3rd

Date of Performance: 18/ 08 / 2026

Subject: Agile Development Methodologies

Subject Code: 25CSH-233

EXPERIMENT – 1.3

6. AIM: Create a Storyboard and Hills for the case discussed in Practical 1 to visualize the user journey and define meaningful, measurable user outcomes.

7. OBJECTIVE:

- To visualize the enhanced cab booking experience through a user-centered Storyboard.
- To represent the complete user journey from the initial situation to the final outcome.
- To formulate meaningful Hill statements based on user needs and Big Idea Vignettes.
- To define clear and measurable user outcomes that guide solution development.
- To use MURAL for collaborative creation and visualization of storyboards and Hills.

8. SOFTWARE REQUIREMENTS:

☐ MURAL

9. THEORY

A. Storyboard Creation

1. Revisit the insights and Big Idea Vignettes developed during Practical 1 and identify important improvements that can enhance the cab booking experience.
2. Identify one or more core user journeys that can be improved through design, such as booking a cab during peak hours, ensuring safety during late-night travel, or confirming the fare before booking.
3. Create a sequence of 6–8 frames in MURAL using sketches, icons, images, or sticky notes.
4. Each frame should represent an important stage of the user's journey, including:
 - The context or situation
 - The action performed by the user
 - The system's response
 - The resulting user emotion
5. Arrange the frames in a logical sequence so that the complete story can be easily understood from the user's perspective.
6. Label each frame clearly and ensure that the visual narrative represents the user's experience from beginning to end.

7. Add the completed Storyboard to the MURAL board under the appropriate section.

Example Storyboard:

Frame 1: User opens the cab booking app in a hurry.

Frame 2: App displays real-time fare and estimated waiting time.

Frame 3: User selects the safety filter to choose a verified driver.

Frame 4: App assigns a verified driver within a short time.

Frame 5: User tracks the cab in real time and feels reassured.

Frame 6: User reaches the destination safely.

Frame 7: Ride is completed and the app asks for feedback.

Frame 8: User provides feedback and feels satisfied with the improved experience.

	Who	What	Wow
	Who are your users? Make it clear who you aim to serve—and who you don't.	What's the need they're trying to meet? Turn user needs into project goals.	How will you differentiate from competitors? How will you measure success?
Examples	A GMU-based sales leader can assemble an agile response team in under 24 hours without management involvement. IBM Connections, 2012 It should take no more than 30 minutes for a developer to build and run an app using IBM and 3rd party APIs. IBM Bluemix, 2014		

B. Hills Creation

1. Understand the concept of Hills, which are statements of intent that focus on meaningful user outcomes rather than only describing system features.
2. Revisit the Need Statements and Big Idea Vignettes created in the previous practical.
3. Convert the identified user goals into clear and actionable Hill statements.
4. Each Hill should:
 - Identify the user/persona
 - Describe what the user wants to achieve
 - Define the desired outcome in a measurable or meaningful way
5. Create suitable Hill statements for different user personas.
6. Add the Hills to a dedicated section in MURAL using user persona images, tags, or sticky notes for better understanding.

Example Hill Statements:

- Ravi, a daily office commuter, can pre-book a ride with real-time fare locking and avoid last-minute cancellations.
 - Sana, a female solo traveler, can feel secure by viewing driver safety badges and route risk information before confirming a ride.
 - Aman, a budget-conscious student, can receive fare suggestions for different time slots and choose a suitable time to avoid surge pricing.
7. Review all Hill statements and ensure that they focus on user outcomes and clearly communicate the desired design goal.



Result:

A visual Storyboard illustrating the enhanced cab booking experience was created, capturing the user's journey step-by-step from opening the application to completing the ride and providing feedback. In parallel, meaningful Hill statements were developed to define the desired user outcomes in an actionable and measurable way. These deliverables provide a clear foundation for user-centric solution design and further development.

Learning Outcomes:

1. Learned how to visually represent user interactions and experiences using Storyboards.
2. Understood how to convert user needs and ideas into meaningful Hill statements.
3. Developed the ability to define user outcomes in a clear and measurable manner.
4. Learned how to communicate design intent effectively from the user's perspective.
5. Practiced using MURAL for collaborative creation of user journey narratives and design outcomes.